

CONTINUAL LEARNING AS SCHRÖDINGER BRIDGE TRANSPORT BETWEEN POSTERIORIS

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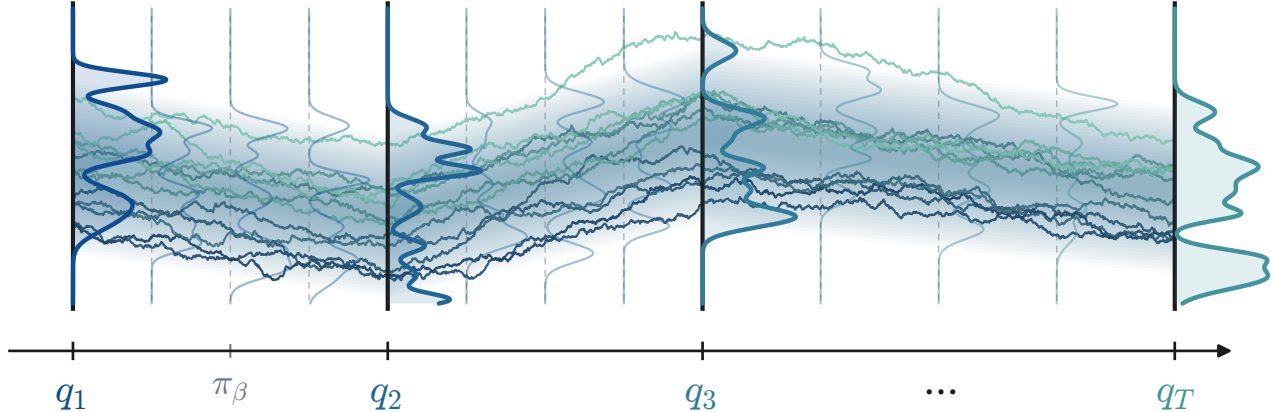


Fig. 1. Continual learning as a chain of $T - 1$ Schrödinger bridges between sequential posteriors $q_t = p(\theta \mid \mathcal{D}_{1:t})$. Bold verticals are task boundaries, dashed verticals are intermediate marginals π_β of one bridge, and the shaded tube is the marginal envelope. Each path is one network of the population, coloured by where it started: the same networks are transported across every boundary rather than re-fitted, and the bridge determines which start is carried to which end. Densities are schematic.

ABSTRACT

Continual learning without replay is usually posed as constrained optimisation: fit the incoming task while a penalty holds the weights near an earlier solution. We pose each task boundary instead as a transport problem between consecutive posteriors. Sequential Bayes fixes both endpoints but not the path. We show that the geometric path between them has a score that splits exactly into a tempered likelihood gradient and a Fisher-weighted prior force, and that annealed Langevin dynamics along it, the natural reference process, lags the path at any finite speed. The minimal correction is a Schrödinger bridge. We transport a population of networks across each boundary and learn the bridge drift by iterative Markovian fitting over per-neuron weight tokens, so that one drift, conditioned on the local likelihood and prior forces, is amortised across the task stream. With one shared head and no stored examples the method reaches 91.0% average accuracy on 10-task Permuted MNIST, against 84%, 86% and 90% reported for EWC, SI and VCL with the same network, and the learned drift carries later boundaries at a quarter of the reference cost for a 0.2-point loss.

Index Terms— Continual learning, Schrödinger bridge, sequential Bayesian inference, optimal transport, catastrophic forgetting

1. INTRODUCTION

A continual learner receives tasks $\mathcal{D}_1, \dots, \mathcal{D}_T$ in sequence and must stay accurate on all of them without revisiting earlier data. The con-

straint is increasingly practical in speech and audio, where deployed recognisers must absorb new domains and speakers while storage or privacy rules forbid keeping the original recordings [1, 2]. A network fitted naively to each new task overwrites what it learned before [3].

Replay-free methods differ in what they carry across a task boundary. Regularisation methods such as EWC [4] and SI [5] keep a point estimate and a diagonal importance vector and add a quadratic penalty. Bayesian methods such as VCL [6] keep an approximate posterior and reuse it as the next prior. Functional methods such as LwF [7] constrain outputs instead of weights. All treat the boundary *locally*: old knowledge enters once, as a fixed quadratic or a factorised Gaussian at the new task’s loss. None models the boundary as what sequential Bayes says it is, the movement of one distribution into another.

Schrödinger bridges (SBs) have recently displaced diffusion models in several speech tasks for a closely related reason. A diffusion model defines a *data-to-noise* process and learns to reverse it from a Gaussian; an SB defines a *data-to-data* process, so generation starts from an informative distribution. This has improved generative speech enhancement [8, 9, 10] and text-to-speech [11], and SBs from a Gaussian to a posterior have been used for Bayesian computation [12]. Continual learning is a data-to-data problem of exactly this kind: sequential Bayes supplies two informative distributions, the posteriors before and after a task, and neither is noise.

We turn this observation into a method, *Posterior Schrödinger Bridges* (PSB), with three contributions. (i) We identify the geometric path between consecutive posteriors as the transport path,

show that its score is exactly a tempered likelihood gradient plus a Fisher-weighted prior force, and show that the annealed Langevin reference along it lags in finite time, which is the defect an SB corrects (Sec. 3.1). (ii) We learn the bridge by iterative Markovian fitting (IMF) on per-neuron weight tokens, which makes regression in weight space well posed and amortises one drift across boundaries (Sec. 3.3). (iii) On 10-task Permuted MNIST, PSB exceeds published EWC, SI and VCL results with the identical single-head network; a controlled sweep attributes the gain to the path rather than to compute, and the drift cuts per-boundary traversal cost by $4\times$ (Sec. 4).

2. BACKGROUND

Sequential Bayes. Let $\theta \in \mathbb{R}^P$ collect the weights of one network shared by all tasks and let $q_t(\theta) = p(\theta \mid \mathcal{D}_{1:t})$. If the datasets are conditionally independent given θ ,

$$q_{t+1}(\theta) \propto \mathcal{L}_{t+1}(\theta) q_t(\theta), \quad \mathcal{L}_{t+1}(\theta) = p(\mathcal{D}_{t+1} \mid \theta). \quad (1)$$

Thus q_t is a sufficient summary of the past. The obstacle is representational: written out, $q_t \propto p(\theta) \prod_{i \leq t} \mathcal{L}_i(\theta)$ needs every past dataset to evaluate, so a replay-free learner must replace it with a lossy summary, and because (1) is recursive each boundary’s approximation error becomes the next boundary’s prior error [13].

Schrödinger bridges. Given a reference path measure Q on $[0, 1]$ and marginals π_0, π_1 , the SB problem [14] is

$$P^* = \arg \min_P \text{KL}(P \parallel Q) \quad \text{s.t.} \quad P_0 = \pi_0, P_1 = \pi_1. \quad (2)$$

If Q is a diffusion $d\theta_s = b(\theta_s, s) ds + \sigma dW_s$ and P has drift $b + u$, Girsanov’s theorem gives $\text{KL}(P \parallel Q) = \frac{1}{2\sigma^2} \mathbb{E}_P \int_0^1 \|u_s\|^2 ds$: P^* reaches π_1 with the least control effort added to the reference, and as $\sigma \rightarrow 0$ it becomes an optimal transport problem whose cost is that effort [14]. P^* is the unique measure that is Markov and lies in the *reciprocal class* of Q , the mixtures of Q ’s bridges over some endpoint coupling. IMF [15, 16] finds it by alternating a Markovian projection, which regresses a drift onto paths of the current coupling, with a reciprocal projection, which regenerates the coupling by simulating the fitted process.

3. POSTERIOR SCHRÖDINGER BRIDGES

3.1. The posterior bridge

Representation. We carry K networks $\{\theta_t^{(k)}\}_{k=1}^K$ and a diagonal precision $\Lambda_t \in \mathbb{R}_+^P$, and approximate

$$q_t(\theta) \approx \frac{1}{K} \sum_k \mathcal{N}(\theta; \theta_t^{(k)}, (\lambda \Lambda_t)^{-1}), \quad (3)$$

with λ a prior-strength constant. The particles carry multimodal structure; the shared precision gives every component support in all P directions. After task t , $\Lambda_t = \gamma \Lambda_{t-1} + F_t$, where F_t is the diagonal Fisher of task t averaged over particles and scaled to unit mean, and $\gamma < 1$ discounts older curvature so the prior does not stiffen uniformly over a long stream. F_t is exact per example: a linear layer’s per-example gradient is an outer product $\delta_n a_n^\top$, so $\sum_n (\delta_n a_n^\top)^{\odot 2} = (\delta^{\odot 2})^\top a^{\odot 2}$ is one matrix product per layer.

The path. Across the boundary $t \rightarrow t+1$ define, for $\beta \in [0, 1]$,

$$\pi_\beta(\theta) \propto \mathcal{L}_{t+1}(\theta)^\beta q_t(\theta) = q_t(\theta)^{1-\beta} q_{t+1}(\theta)^\beta, \quad (4)$$

where the equality uses (1). This is the geometric path of annealed importance sampling and SMC samplers [17, 18], and it runs exactly from q_t to q_{t+1} . It tempers the *likelihood* rather than corrupting the *parameters*, so every π_β is a posterior over functioning networks. This matters in weight space: with $K \leq 256$ samples in $P = 89,610$ dimensions, isotropic forward noise puts over 99.99% of its energy outside the span of the samples, where no score can be estimated, so a data-to-noise construction cannot be inverted.

We transport each mixture component separately. Reweighting components by their evidence under $\mathcal{L}_{t+1}^\beta$ would be exact, but with 60,000 examples the weights are degenerate: the effective sample size of 32 particles collapses to 1.4. For particle k the score of the path is

$$\nabla_\theta \log \pi_\beta^{(k)}(\theta) = \underbrace{-\beta \nabla_\theta \ell_{t+1}(\theta)}_{\text{tempered likelihood}} - \underbrace{\lambda \Lambda_t (\theta - \theta_t^{(k)})}_{\text{Fisher-weighted prior}}, \quad (5)$$

with ℓ_{t+1} the mean negative log-likelihood of task $t+1$. The second term is the gradient of the EWC penalty, but here it is balanced against a likelihood whose weight grows with β . Setting (5) to zero,

$$\theta - \theta_t^{(k)} = -\frac{\beta}{\lambda} \Lambda_t^{-1} \nabla_\theta \ell_{t+1}(\theta), \quad (6)$$

so displacement grows with β and is rationed per coordinate by $1/\Lambda_{t,j}$: weights that earlier tasks rely on barely move, weights they left free absorb the new task. Since (6) is nonlinear with many roots, sweeping β upward from 0 is a numerical continuation: it follows the solution branch that starts at the old network instead of jumping to an arbitrary minimiser of the new task.

3.2. Reference process and the bridge problem

The SB cost is measured against a reference. Brownian motion, the usual choice, is a poor one here: its cost is Euclidean length, which prices a small move in a critical weight like a large move in an irrelevant one, and its bridges are straight lines through regions where interpolated networks fail. We instead take as reference Q the annealed Langevin diffusion along the path,

$$d\theta_s = \nabla_\theta \log \pi_{\beta(s)}^{(k)}(\theta_s) ds + \sqrt{2\tau} dW_s, \quad \beta(s): 0 \nearrow 1. \quad (7)$$

Its drift is (5), so moving a weight against its prior costs control effort in proportion to $\Lambda_{t,j}$: the Fisher metric enters through the reference instead of being imposed.

At temperature $\tau = 1$ and run infinitely slowly, (7) keeps $\pi_{\beta(s)}$ as its marginal at every s and ends at q_{t+1} . At finite speed it *lags*: the population is still relaxing toward $\pi_{\beta(s)}$ when β has moved on, and it ends short of q_{t+1} . The posterior bridge

$$P_t^* = \arg \min_P \text{KL}(P \parallel Q) \quad \text{s.t.} \quad P_0 = q_t, P_1 = q_{t+1} \quad (8)$$

is the smallest correction to (7) that removes the lag. By Doob’s h -transform its drift is the reference drift plus a correction, so the learned drift need not be exact: we deploy it alongside a few reference moves that correct its residual error.

Discretisation. Each boundary is split into M stages with $\beta_m = m/M$, and each stage applies n moves of the $\tau \rightarrow 0$ limit of (7), split into its two terms:

$$\theta \leftarrow \theta - \beta_m \eta \mathcal{P}(g), \quad \theta \leftarrow \frac{\theta + w \odot \theta_t^{(k)}}{1 + w}, \quad w = \lambda \eta \Lambda_t, \quad (9)$$

where g is a minibatch gradient of ℓ_{t+1} , $\mathcal{P}$ the Adam preconditioner, and division is elementwise. The second step is the exact proximal

operator of the prior term. Adding the prior to the loss instead would route its gradient through $\mathcal{P}$, which divides each coordinate by its own gradient scale and cancels exactly the Λ -weighting the prior exists to impose; applied proximally the weighting survives and the step is stable for any w . At $\tau \rightarrow 0$ particles track the path rather than sample it, and the population supplies the spread.

Special cases. With $M = 1$ the path collapses to $\beta = 1$ and (9) is EWC with a decayed Fisher, applied proximally and run on K networks; with $K = 1$ as well it is a single EWC-style learner. PSB therefore differs from this family only in transporting along π_β , and Fig. 2 isolates exactly that difference.

3.3. Learning the bridge by IMF over neuron tokens

Tokens. Regression over $\theta \in \mathbb{R}^{89,610}$ from $K = 128$ networks has 1.4×10^{-3} samples per dimension. A network is also a set of neurons: a layer with n_ℓ units and fan-in d_ℓ yields n_ℓ tokens $z \in \mathbb{R}^{d_\ell+1}$ (incoming weights and bias). The map is invertible, and tokens of a layer share one regression, giving 16, 127 and 13 samples per dimension for our three layers from a single snapshot of the population.

Drift. Per layer we fit $d_\phi(z, g, a, \beta)$, the displacement Δz of a token over one stage given its value z , its likelihood-gradient token g , its prior-force token a (the two terms of (5)), and β :

$$\min_{\phi} \mathbb{E} \|d_\phi(z, g, a, \beta) - \Delta z\|^2. \quad (10)$$

This is IMF’s Markovian projection. Conditioning on forces rather than on task identity makes d_ϕ a rule, how far to move under this much evidence against this much stiffness, so a drift fitted on early boundaries applies to later ones. d_ϕ is an MLP on RMS-normalised inputs, their log-scales and Fourier features of β , with a zero-initialised output layer so that an unfitted drift leaves the reference unchanged.

IMF along the stream. The first W boundaries use the reference ($n = n_{\text{ref}}$) and record per-stage displacements. Each IMF round then fits a forward half-bridge d_ϕ by (10) and a backward half-bridge d_ψ on reversed pairs ($z + \Delta z \mapsto -\Delta z$), and discards the recorded coupling. The next boundary is traversed with the drift, one drift step per stage followed by $n_{\text{dep}} \ll n_{\text{ref}}$ reference moves, and its displacements become the new coupling: this re-simulation is the reciprocal projection. At the bridge the half-bridges invert each other, so the cycle residual

$$C = \mathbb{E} \|d_\phi(z) + d_\psi(z + d_\phi(z))\|^2 / \mathbb{E} \|d_\phi(z)\|^2 \quad (11)$$

measures the distance from the fixed point. The task stream thus supplies IMF’s iterations; after R rounds the drift is frozen (Algorithm 1).

4. EXPERIMENTS

Setup. *Permuted MNIST* [3]: 10 tasks, each a fixed pixel permutation, the first being the identity. *Rotated MNIST*: 10 tasks rotated by $0^\circ, 20^\circ, \dots, 180^\circ$. *Split MNIST* [5]: 5 tasks on digit pairs, evaluated domain-incremental (a shared binary head) and class-incremental (a shared 10-way head) [19]. The network is a 784–100–100 ReLU MLP with one shared head and no task-specific parameters ($P = 89,610$ for Permuted MNIST), the network behind the published Permuted MNIST results for EWC, SI, LP and VCL [6]. We use $K = 128$; Adam with $\eta = 10^{-3}$ and batch 64; 3,000 steps on task 1; $M = 25$ stages with $n_{\text{ref}} = 100$; $\lambda = 4$, $\gamma = 0.8$; and Fisher

Algorithm 1 Posterior Schrödinger Bridges (PSB)

Require: tasks $\mathcal{D}_{1:T}$; $K, M, n_{\text{ref}}, n_{\text{dep}}, W, R, \lambda, \gamma$

- 1: Train K networks on $\mathcal{D}_1$ independently; $\Lambda_1 \leftarrow F_1$
- 2: **for** $t = 1, \dots, T - 1$ **do**
- 3: Freeze centres $\theta_t^{(k)}$; $n \leftarrow n_{\text{dep}}$ if d_ϕ fitted, else n_{ref}
- 4: **for** $m = 1, \dots, M$ **do** $\triangleright \beta_m = m/M$
- 5: **if** d_ϕ fitted: move every token by $d_\phi(z, g, a, \beta_m)$
- 6: Apply n moves (9); record $(z, g, a, \beta_m, \Delta z)$
- 7: **if** $W \leq t < W + R$ **then** $\triangleright$ one IMF round
- 8: Fit d_ϕ, d_ψ by (10); clear the record
- 9: $\Lambda_{t+1} \leftarrow \gamma \Lambda_t + F_{t+1}$
- 10: **return** ensemble $\frac{1}{K} \sum_k p(y | x, \theta_t^{(k)})$

Table 1. Ten-task Permuted MNIST, single head, no stored examples. Published results use the same 784–100–100–10 network [6]; coreset variants of VCL store examples and are excluded.

Method	Avg. acc.	Forgetting	Acquisition
LP [6]	0.82	–	–
EWC [4]	0.84	–	–
SI [5]	0.86	–	–
VCL [6]	0.90	–	–
PSB	0.910 ± 0.001	0.035	0.941
PSB, 4,000 moves [†]	0.914	0.034	0.945

from 20,000 examples. The drift has two hidden layers of width 512, is trained for 3,000 AdamW steps on tokens from 8 particles, and is deployed with step scale 0.3, $W = R = 3$ and $n_{\text{dep}} = 25$. With $a_{i,j}$ the accuracy on task j after task i , we report average accuracy $\frac{1}{T} \sum_j a_{T,j}$, forgetting $\frac{1}{T-1} \sum_{j < T} (\max_{i \leq j} a_{i,j} - a_{T,j})$, and acquisition $\frac{1}{T} \sum_j a_{j,j}$, which separates failing to learn from failing to retain. Results are means over five seeds; rows marked [†] are single runs.

Comparison. Table 1 compares PSB with published results for the same network. PSB reaches 0.910 with seed spread 0.001, above VCL’s 0.90, and a longer traversal of 4,000 moves per boundary reaches 0.914. Forgetting is low and uniform: seed-averaged accuracies on tasks 1–10 after the full stream are 0.937, 0.913, 0.913, 0.913, 0.872, 0.899, 0.905, 0.916, 0.908 and 0.922, with the oldest task the most accurate.

The path, not the budget. Because β scales the data step, the bridge could be mistaken for a learning-rate warm-up. Fig. 2 separates the two. As the budget grows from 2,500 to 20,000 moves per boundary, traversal along π_β improves from 0.901 to 0.912, while the same moves with β fixed at 1 *degrade* from 0.889 to 0.879. Off the path, extra moves raise acquisition (0.954 $\rightarrow$ 0.958) but raise forgetting faster (0.073 $\rightarrow$ 0.088), buying the new task at the old tasks’ expense; on the path, acquisition rises (0.949 $\rightarrow$ 0.956) while forgetting falls (0.053 $\rightarrow$ 0.049). Under identical budgets the curves move in opposite directions, so the gain comes from the path. That the blue curve has not saturated at 20,000 moves is the finite-time lag of Sec. 3.2, the quantity the bridge exists to remove.

Learning the bridge. Table 2 (top) shows IMF converging. After one round, which is bridge matching, the half-bridges explain 29% and 17% of the stage displacement and do not invert each other ($C = 0.908$). After a second round, fitted on the coupling that the first drift generated, they explain 62% and 67% and nearly invert ($C = 0.114$). Closing the cycle does not raise final accuracy: at

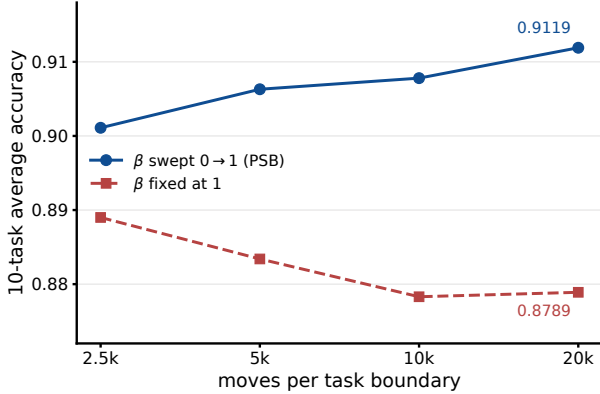


Fig. 2. The path, not the budget. Moves (9) with β swept $0 \rightarrow 1$ (blue) versus the same moves with β fixed at 1 (red), as the number of moves per boundary grows; $K = 32$, all else matched.

Table 2. Learning the bridge ($K = 128$). Top: IMF rounds; explained variance of the forward and backward half-bridges and cycle residual (11). Bottom: kernel moves per boundary once the drift is deployed.

IMF round	fwd. R^2	bwd. R^2	cycle $\mathcal{C}$
1 (bridge matching)	0.286	0.168	0.908
2	0.624	0.666	0.114
Traversal	moves	Avg. acc.	Forgetting
Reference	2,500	0.910	0.035
Drift [†]	2,500	0.910	0.035
Drift [†]	625	0.908	0.028

$K = 32$, $\mathcal{C}$ falls from 0.274 to 0.104 while accuracy moves from 0.9015 to 0.9017, and at $K = 128$ it moves from 0.907 to 0.904. The endpoints of each bridge are fixed by (4); the bridge governs how cheaply they are connected, not where they lie. This matches the view of Kessler et al. [13] that continual learning under sequential Bayes is limited by how well each posterior is represented. The bottom of Table 2 measures that cost. At the reference budget the drift matches the reference (0.9103 vs. 0.9099); with a quarter of the moves it reaches 0.908. Since the first $W = 3$ boundaries use the reference, total moves over this 10-task stream halve, and the saving approaches $4\times$ as the stream grows.

Other benchmarks. Table 3 reports Rotated and domain-incremental Split MNIST. Accuracy rises monotonically with λ on both and has not peaked at $\lambda = 16$, so the best setting lies at or beyond the values tested. Both are markedly harder for PSB than Permuted MNIST, with forgetting ten times higher. We attribute this to the diagonal precision: different permutations load largely different input weights, which a diagonal Λ can protect separately, whereas rotations and digit pairs reuse the same pixels with conflicting targets, which a coordinate-wise prior cannot arbitrate. Structured precisions [20] are the natural extension and slot into (5) unchanged. Class-incremental Split MNIST reaches 0.224 ± 0.009 against a chance level of 0.20: with two classes per task nothing trains the absent classes’ logits, an output-layer failure that no weight-space prior addresses and that we leave out of scope.

Limitations. Baselines are quoted from [6] for the same network; training schedules and tuning may differ, and we have not

Table 3. Average accuracy against prior strength λ , five seeds (s.d. ≤ 0.009). Forgetting at $\lambda = 16$ is 0.37 and 0.41.

λ	1	2	4	8	16
Rotated	–	0.540	0.551	0.568	0.593
Split, domain	0.636	–	0.649	–	0.662

re-run them. PSB predicts with a K -member ensemble, and ensembling alone improves accuracy [21]. The $\beta=1$ curve of Fig. 2 is an ensemble of the same size with the same prior and trails PSB at every budget, but it is a $K = 32$ proximal variant; a standard EWC ensemble at $K = 128$ remains the cleaner control. Persistent state is K networks, Λ_t and a 2.9M-parameter drift, independent of T but larger than VCL’s $2P$. We work at $\tau \rightarrow 0$, so particles track rather than sample each π_β .

5. CONCLUSION

Posing the continual-learning task boundary as transport between consecutive posteriors identifies the geometric path as the route and the Schrödinger bridge as the way to traverse it in finite time. Tempering the likelihood rather than corrupting the weights keeps every intermediate distribution a posterior over working networks, which is what makes a bridge viable in weight space, and the score of that path recovers the likelihood and Fisher-prior forces that replay-free methods use, now tied together by a schedule with a precise meaning. A drift learned by iterative Markovian fitting over neuron tokens converges toward the bridge and carries later boundaries at a fraction of the cost. Structured precisions and bridges between function-space posteriors are the natural next steps.

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